

MemDistill: Diagnosing Why Distilling RL Agent Memory Fails

Anonymous ACL submission

Abstract

Distilling RL-trained LLM agents into smaller students via supervised fine-tuning (SFT) fails catastrophically, yet the failure is *silent* at the format level: students pass tool-call syntax, balanced-XML, and query-similarity checks while retaining $<6\%$ of teacher task performance. We present a diagnostic protocol—collapse characterization, weight-geometry analysis, and causal intervention—and demonstrate it on MEM1, a 7B RL agent for multi-objective retrieval-augmented QA distilled into Qwen2.5-3B and, to test generality, a cross-family Llama-3.2-1B student. Providing oracle retrieval at inference recovers `em.partial` from 0.025 to 0.498 at $N = 8$ (+47.3pp, $p < 0.0001$), causally attributing $\sim 86\text{--}88\%$ of the teacher-student gap to behavioral mismatch and *at most* $\sim 12\text{--}14\%$ to capacity (quantified via a 7B full-rank SFT). BM25 retrieval recovers +6–12pp as a deployable mitigation. The entire silent-failure signature—format-level indicators near 1.0, content-level metrics (action-seq ≈ 0.606 , BLEU ≈ 1.15 , ROUGE-L ≈ 0.21) near-floor, and oracle recovery—replicates on Llama-1B, elevating the finding from a case study to a systematic SFT-distillation failure mode.

1 Introduction

SFT distillation of an RL-trained agent can appear to succeed by format-level tool-call metrics—and still fail catastrophically in deployment. We distill MEM1 (Zhou et al., 2025), a Qwen2.5-7B retrieval-augmented QA agent trained with PPO via veRL (Sheng et al., 2024), into a Qwen2.5-3B student using 1,118 teacher trajectories, and we replicate the entire pipeline with a Llama-3.2-1B student (cross-family, $\sim 1/3$ the Qwen-3B student size) to test whether the failure is backbone-specific or systematic (§3.6). On both backbones, format-level tool-call indicators rate the student at 1.00

(tool-call syntax, balanced XML tags, non-empty queries; Qwen query Jaccard = 0.95), while content-level metrics reveal the collapse: BLEU = 1.14/1.16 on sacrebleu’s 0–100 scale, ROUGE-L = 0.21/0.21, action-sequence accuracy ≈ 0.606 for both students, and `em.partial` at $N = 8$ retaining only 0.025/0.014 of teacher capability (full per-metric breakdown in Appendix H). These content warnings are easy to dismiss as cosmetic noise because the student produces teacher-like *structure*—a blind spot mirroring the “syntax-only” tool-call evaluations critiqued by Jhandi et al. (2025) (§4).

It did not succeed on either backbone. The Qwen-3B student achieves `em.partial` = 0.025 at $N = 8$ simultaneous objectives—a 94.5% drop from the teacher’s 0.455—with 83% of outputs truncated by hallucinated search traces. The Llama-1B student reaches `em.partial` = 0.014 at $N = 8$ with 86% truncation—a near-identical collapse despite the change in both family and parameter count. The failure is *silent* at the format/syntax layer on both backbones: the format-level checks all pass while action-sequence accuracy at the token level is only ≈ 0.606 for both students¹. The collapse surfaces only when search calls must return real results, i.e., at pipeline deployment. This is the central danger: an invisible failure—reproducible across backbones—that passes format-level tool-call quality gates while content-level warning signs are overridden by teacher-like structure.

Why do richer training signals not help? DPO on matched preference pairs yields +1.0pp ($p = 0.163$)—preference learning fails because the problem is not what the student prefers to generate. Scaling from 3B to 7B SFT closes only

¹Near-identical point estimates reflect floor-level action-prefix bunching (both students begin with the same “search” action); per-sample dispersion and bootstrap 95% CIs in Appendix I.

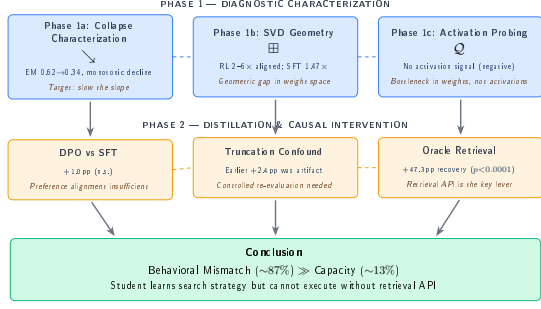


Figure 1: Overview of the MEMDISTILL diagnostic framework. Phase 1 (top) progressively characterizes the distillation failure through collapse analysis, SVD weight geometry, and activation probing. Phase 2 (middle) tests interventions: DPO preference learning, truncation confound correction, and oracle retrieval. The combined evidence (bottom) attributes $\sim 87\%$ of the teacher–student gap to behavioral mismatch.

$\sim 12\text{--}14\%$ of the teacher–student gap (§3.2), ruling out capacity as the primary bottleneck.

We introduce a diagnostic protocol that distinguishes *strategy acquisition* from *execution capability*. The key causal test: providing oracle retrieval at inference recovers `em_partial` from 0.025 to 0.498 at $N=8$ (+47.3pp, $p < 0.0001$), with truncation dropping from 83% to 0%. Even a simple BM25 retrieval system recovers +6–12pp. The protocol proceeds by elimination—collapse characterization and weight geometry analysis (§2), causal intervention (§3)—and localizes the failure before deployment (Figure 1).

Contributions.

- A **silent failure mode** in agentic SFT distillation: format-level indicators (tool-call syntax, balanced XML tags, query similarity) give high scores while task performance retains $< 6\%$ of teacher capability.
- A **diagnostic protocol** (collapse characterization $\rightarrow$ weight geometry $\rightarrow$ causal intervention) that localizes the failure to execution–environment mismatch before pipeline deployment, applicable to any agentic distillation where the teacher’s capability depends on tool execution.
- **Causal evidence and quantification:** oracle retrieval +47.3pp ($p < 0.0001$); capacity accounts for $\sim 12\text{--}14\%$ of the gap; behavioral mismatch accounts for $\sim 86\text{--}88\%$; DPO is an informative negative (power > 0.99 at 5pp).

- A **deployable mitigation:** BM25 retrieval recovers +6–12pp with no additional training, immediately deployable as a sidecar component.

2 Diagnostic Framework

The diagnostic protocol proceeds in two phases. Phase 1 (§2) localises the failure through three progressively narrower analyses; Phase 2 (§3) tests causal interventions distinguishing strategy acquisition from execution capability.

Phase 1: collapse $\rightarrow$ geometry $\rightarrow$ activations.

We evaluate the MEM1 RL policy (Qwen2.5-7B + PPO, trained via `veRL` (Sheng et al., 2024)) and the Qwen2.5-7B base on a multi-objective RAG QA task from FlashRAG (Jin et al., 2025b); each prompt contains N questions ($N \in \{1, 2, 4, 8, 12, 16\}$) and the agent runs `think` $\rightarrow$ `search` $\rightarrow$ `information` $\rightarrow$ `answer` with per-objective normalized EM (`em_partial`). Three signals together localise the gap. **(1a) Collapse curve.** MEM1’s `em_partial` decreases monotonically from 0.62 at $N=1$ to 0.341 at $N=16$ (a 45% drop, $\text{EM} \approx 0.62 - 0.018N$, $R^2 = 0.984$, Cohen’s $d = 0.97$, no inflection point); the base model collapses immediately (0.30 at $N=1$, 0.075–0.148 for $N \geq 2$), so the distillation target is slope reduction rather than cliff recovery (Figure 2). **(1b) Weight geometry.** Across all 196 weight matrices of Qwen2.5-7B we measure the alignment ratio $r(k) = \alpha_{\text{obs}}(k)/\alpha_{\text{rand}}(k)$ between fine-tuning deltas $\Delta \mathbf{W} = \mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ and the top- k base singular subspace, with α_{rand} calibrated against $n = 100$ Gaussian matrices matched in shape and variance. RL–base alignment is $2\text{--}6\times$ supra-random (e.g. $6.48\times$ at $k = 64$, $2.78\times$ at $k = 256$), whereas same-architecture full-rank SFT–base recovers only $1.47\times$ at $k = 256$ ($2.92\times$ at $k = 64$); a Wilcoxon signed-rank test gives $p < 1.3 \times 10^{-32}$ at every k and 99% of matrices have $r_{\text{RL}} > r_{\text{SFT}}$ (Figure 3; full per-layer / U–V decomposition in Appendix C). **(1c) Activation probing (negative).** Neither linear probes nor 2-layer MLPs ($n = 300$ prompts, 6 layers, $k \in \{64, 256\}$) recover a per-objective signal from $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$: linear binary accuracy $0.72 < 0.76$ majority-class, MLP $<$ linear in 47/55 cases ($p = 4.83 \times 10^{-11}$), and 94.5% of real $|\rho|$ values fall within a random-projection noise band (Appendix D). The bottleneck thus

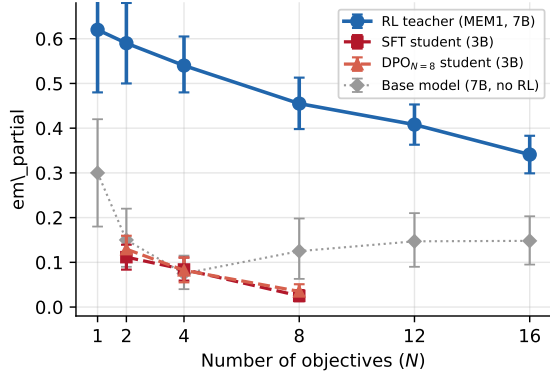


Figure 2: Multi-objective performance collapse. MEM1 degrades monotonically with N ; SFT/DPO students retain $<8\%$ of teacher capability at $N = 8$; the base model collapses immediately. 95% bootstrap CIs (10,000 resamples).

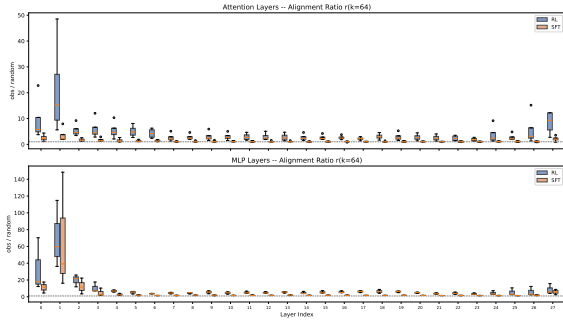


Figure 3: Per-layer SVD alignment ratio $r(k=256)$ for RL-base and SFT-base deltas across all 196 weight matrices. RL alignment exceeds SFT in 99% of layers (Wilcoxon $p < 1.3 \times 10^{-32}$).

lives in weight-space geometry, not activations, motivating the causal interventions of §3.

3 Distillation Attempt and Failure Analysis

Phase 1 narrowed the distillation failure to a geometric gap in weight space that SFT does not close. We now test whether this gap reflects a causal bottleneck—and if so, where it lies. We establish DPO and capacity baselines (§3.2), examine the truncation-behavioral-mismatch link (§3.3), and test the causal hypothesis (§3.4).

3.1 Experimental Design

Training pipeline. We train a pooled SFT baseline and three per- N DPO variants on Qwen2.5-3B (LoRA (Hu et al., 2022) rank 32). SFT uses 1,118 partial-credit trajectories (pooled $N \in \{2, 4, 8\}$; $\text{total_em} \geq 1$) for 3 epochs at $\text{lr} = 2 \times 10^{-4}$. Each DPO variant initialises from

the SFT checkpoint, trains on N -specific preference pairs matched to 176 raw pairs at $\beta = 0.05$, $\text{lr} = 2 \times 10^{-5}$, 1 epoch. Full hyperparameters in Appendix A.

Evaluation. Models are evaluated on held-out splits (hash-based, $\text{test_frac} = 0.25$) for $N \in \{2, 4, 8\}$ with $\text{max_new_tokens} = 4096$ (251/131/125 prompts). We report em_partial , em_full , and truncated_rate ; significance via paired bootstrap (1,000 resamples); pre-registered practical-significance threshold $\Delta \text{em_partial} \geq +5\text{pp}$ at $N = 8$. The matched-sample constraint subsamples DPO to the $N = 4$ bottleneck of 176 raw pairs—a structural consequence of multi-objective collapse rather than a remediable sampling issue (Appendix E).

3.2 DPO and Capacity Ablations

We also evaluate DPO on matched preference pairs as a stronger-than-SFT baseline; the detailed analysis (Appendix E) confirms DPO yields only $+1.0\text{pp}$ ($p = 0.163$, power > 0.99 at 5pp), ruling out preference learning as a fix.

Capacity isolation. A 7B full-rank pooled SFT (same data/inclusion as the 3B LoRA student) consistently outperforms the 3B student (0.171/0.147/0.079 vs. 0.112/0.084/0.025 at $N = 2/4/8$). Defining the capacity contribution as $(\text{em}_{7\text{B}} - \text{em}_{3\text{B}}) / (\text{em}_{\text{teacher}} - \text{em}_{3\text{B}})$, the 7B–3B gap accounts for *at most* 12.3/13.8/12.6% of the teacher–student gap—a stable $\approx 12\text{--}14\%$ upper bound (the 7B uses full-rank fine-tuning, more expressive than rank-32 LoRA, so the training-method advantage is conflated in). Behavioral mismatch (§3.4) therefore accounts for the remaining $\approx 86\text{--}88\%$.

3.3 Truncation and Behavioral Mismatch

We identified and controlled for a truncation artefact that inflated early DPO signal: without retrieval, both students hallucinate `<search>/<information>` segments that consume $> 80\%$ of the generation budget, causing $\geq 83\%$ output truncation at $N = 8$ (truncation scales monotonically as $30.7\% \rightarrow 55.7\% \rightarrow 83.2\%$ for $N = 2/4/8$, with DPO showing the same pattern). A suppression ablation (Appendix G) confirms truncation is the surface manifestation, not the root cause, of behavioral mismatch.

	$N=2$	$N=4$	$N=8$
SFT (no API)	0.112	0.084	0.025
SFT + oracle API	0.530	0.521	0.498
Δ	+41.8pp	+43.7pp	+47.3pp
p (bootstrap)	<0.0001	<0.0001	<0.0001
RL teacher [†]	0.59	0.54	0.455

Table 1: Effect of providing oracle retrieval results at inference time. With real search results injected, the SFT student recovers from near-zero to near-teacher performance. Paired bootstrap, 10,000 resamples. [†]Teacher evaluated on a different prompt set; direct comparison is not valid.

	$N=2$	$N=4$	$N=8$
SFT (no API)	0.112	0.084	0.025
SFT + BM25	0.203	0.147	0.147
SFT + oracle API	0.530	0.521	0.498
RL teacher	0.59	0.54	0.455

Table 2: em_{partial} under different retrieval conditions. BM25 retrieval recovers a meaningful fraction of the oracle gain (+9.2pp at $N=2$, +12.2pp at $N=8$) and eliminates truncation entirely (0% across all N). The remaining gap to oracle reflects retrieval quality, not student strategy.

3.4 Causal Verification: Oracle Retrieval

If the failure stems from executing search procedures without a retrieval backend, providing the backend should recover performance. We test this by intercepting the student’s `<search>` outputs at inference time and injecting oracle retrieval results from the RL teacher’s trajectories.

Table 1 shows oracle retrieval transforms student performance: em_{partial} jumps $0.025 \rightarrow 0.498$ at $N=8$ (+47.3pp, $p < 0.0001$, 95% CI [43.8, 50.8]pp); truncation drops 83.2% $\rightarrow$ 0%. This is causal evidence that behavioral mismatch—not capacity—drives the failure: the student has learned a viable search strategy but lacks the execution environment.

Realistic retrieval: BM25. Replacing oracle with BM25 (top-4 passages from the same corpus) recovers 14–26% of the oracle gain: em_{partial} improves $0.025 \rightarrow 0.147$ at $N=8$ (+12.2pp; Table 2, Figure 4), with truncation dropping to 0% across all N . The BM25–oracle gap reflects retrieval quality, not student capability: the student’s queries match the teacher’s (Jaccard = 0.95).

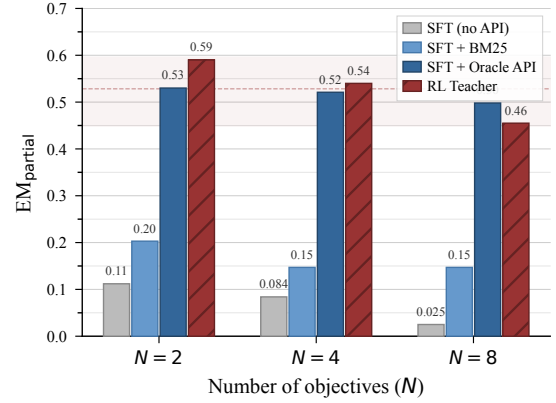


Figure 4: em_{partial} under different retrieval conditions. Providing oracle retrieval results at inference time recovers the SFT student from near-zero to near-teacher performance across all N , while even a standard BM25 backend yields meaningful gains. The RL teacher (hatched) is shown for reference.

Query fidelity. The student averages exactly 1.0 search per prompt across 507 test cases—matching the teacher’s 1.04/1.01/1.06 at $N=2/4/8$ (the prompt prefix already includes the first retrieval result)—and its queries match the teacher’s at mean Jaccard 0.95 (median 1.0). SFT faithfully transfers both the search pattern and the query formulation; the bottleneck is purely the absence of a retrieval backend. Despite recovering em_{partial} to near-teacher levels, em_{full} remains low (0.008 at $N=8$) due to per-objective $\approx 50\%$ independence stacking ($0.5^N \approx 0.4\%$ at $N=8$).

3.5 Synthesis

The evidence converges on three points: (i) **preference learning does not close the geometric gap**—DPO remains within 1pp of SFT at $N=8$ despite richer training signal, against the 2–6 $\times$ vs. 1.47 $\times$ supra-random alignment gap (§2); (ii) **the student imitates a strategy it cannot execute**—oracle retrieval recovers em_{partial} $0.025 \rightarrow 0.498$ at $N=8$ (+47.3pp, $p < 0.0001$; §3.4), confirming a viable search strategy (Jaccard 0.95) but no execution environment; (iii) **truncation distorts the comparison**—after controlling for budget, the DPO advantage shrinks from +2.4pp ($p < 0.001$) to +1.0pp ($p = 0.163$). Capacity (§3.2) accounts for ≈ 12 –14% of the gap, behavioral mismatch the remaining ≈ 86 –88% (Figure 5); these are approximate, possibly non-additive decomposi-

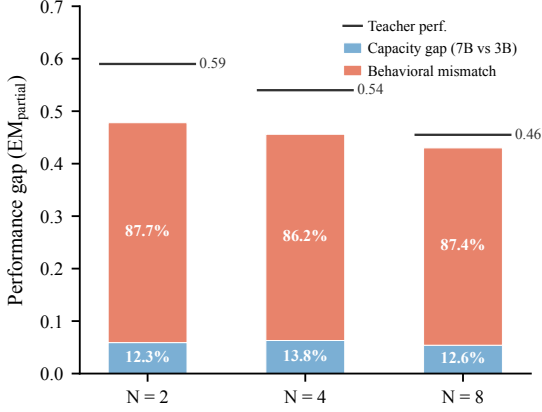


Figure 5: Teacher-student gap decomposition. Capacity (7B vs. 3B SFT) $\approx 12\text{--}14\%$; behavioral mismatch $\approx 86\text{--}88\%$.

Backbone	Condition	$N=2$	$N=4$	$N=8$
Qwen-3B	No API	0.112	0.084	0.025
	+ BM25	0.203	0.147	0.147
	+ Oracle	0.530	0.521	0.498
Llama-1B	No API	0.074	0.063	0.014
	+ BM25	0.136	0.094	0.066
	+ Oracle	0.410	0.401	0.351
RL teacher (7B)	—	0.59	0.54	0.455

Table 3: Cross-backbone replication of the diagnostic framework (em_partial). The Llama-1B student reproduces the Qwen-3B silent-failure signature: near-floor no-API performance, partial BM25 recovery, and large oracle recovery (truncation drops from 30–86% to 0% across all N). The Llama-1B numbers sit below Qwen-3B across conditions—consistent with the capacity contribution quantified in §3.2—but the *diagnostic shape* is preserved.

tions.

3.6 Cross-Backbone Replication

To test whether the diagnosis generalises beyond Qwen2.5-3B, we re-run the entire pipeline—same trajectories, splits, LoRA rank-32 recipe, and eval protocol—on Llama-3.2-1B, a cross-family backbone at $\sim 1/3$ the size.

Table 3 shows every diagnostic signal reappears on Llama-1B: (i) truncation scales monotonically ($29.9\% \rightarrow 51.2\% \rightarrow 85.6\%$ vs. Qwen’s $30.7\% \rightarrow 55.7\% \rightarrow 83.2\%$); (ii) oracle retrieval drops truncation to 0% at every N with a +33.7pp gain at $N=8$ (vs. +47.3pp for Qwen, both large and monotonic in N); (iii) BM25 partially recovers performance and also collapses truncation. Surface metrics at $N=8$ are backbone-agnostic to within noise: tool-

call syntax = 1.00, balanced XML = 1.00, non-empty queries = 1.00, action-sequence accuracy = 0.606, BLEU = 1.16, ROUGE-L = 0.21 (Appendix H); the smaller cross-family backbone lowers the absolute ceiling but preserves the diagnostic shape. The transferable lesson is the protocol, not the specific 3B model it was developed on; near-identical surface estimates reflect discrete action-prefix and floor-level BLEU/ROUGE bunching, with overlapping bootstrap CIs (Appendix I).

4 Related Work

RL-trained LLM agents. RL fine-tuning endows LLMs with procedural capabilities SFT does not produce (Zhou et al., 2025, 2024a; Liu et al., 2024; Yao et al., 2023; Zeng et al., 2024). We investigate why these RL-acquired strategies resist distillation.

Knowledge distillation for LLMs. Standard methods (Hsieh et al., 2023; Gu et al., 2024; Jiang et al., 2023) target single-turn transfer; policy distillation and imitation learning (Rusu et al., 2016; Czarnecki et al., 2019; Ross et al., 2011) handle procedural knowledge in low-dimensional action spaces. Our setting differs: the teacher’s competence arises from a *multi-turn procedural policy* in the high-dimensional token space of LLMs.

SFT-RL alignment gaps and weight-space analysis. The “alignment tax” (Askell et al., 2021) documents RLHF degrading pre-training capabilities. Jin et al. (2025a) show SFT induces “hard alignment” while RL produces “soft alignment”; Aghajanyan et al. (2021) demonstrate fine-tuning operates in low intrinsic dimension. PiSSA (Meng et al., 2024) initializes LoRA from principal singular components; our SVD alignment ratio measures the overlap of *fine-tuning deltas* with the base model’s principal subspace. We extend these by revealing quantitative gaps ($2\text{--}6\times$ for RL vs. $1.5\times$ for SFT) not captured by prior metrics. Extended related work on multi-objective RL and preference learning is in Appendix F.

The evaluation blind spot in agentic distillation. Recent work increasingly distills tool-using capabilities from large models into smaller ones, but evaluation practices vary widely in whether they test actual tool execution. Several

works evaluate only format correctness: Jhandi et al. (2025) use ChatGPT-as-judge to score tool-call syntax without executing calls; Yin et al. (2025) evaluate against BFCL format matching, which measures argument-level accuracy but not end-to-end execution; AgentKD (ModelScope Team, 2025) trains on virtual tool-use tasks with pre-defined return values. These evaluations implicitly assume that format accuracy entails execution capability. Our results show this assumption can fail catastrophically: Jaccard = 0.95 (near-perfect format) coexists with em_partial = 0.025 (near-zero task performance). Works that provide real tool access at test time (Kang et al., 2025; Liu et al., 2025) produce more robust evaluations but still lack a with/without tool-backend ablation that would reveal whether the student genuinely depends on tool execution. To our knowledge, no prior agentic distillation work reports an oracle retrieval ablation that directly compares student performance with and without the tool backend on the same inputs—the diagnostic that directly quantifies the execution gap. We propose this ablation as a standard component of agentic distillation evaluation.

5 Conclusion

Agentic SFT distillation can produce students that pass format-level, tool-call structural checks—perfect tool-call syntax, balanced XML tags, near-perfect query similarity—yet retain less than 6% of the teacher’s task performance, and we show this collapse replicates across two backbones and two parameter scales (Qwen2.5-3B and Llama-3.2-1B, spanning two model families and a $\sim 3\times$ size difference). We have proposed a diagnostic protocol to detect this silent failure mode before pipeline deployment. The protocol proceeds by elimination: characterize the collapse shape, analyze weight geometry for geometric gaps between RL and SFT, and apply causal interventions (oracle and BM25 retrieval) that distinguish strategy acquisition from execution capability. Because the protocol requires only inference-time ablations, it is transferable to any agentic distillation setting where the teacher’s competence depends on tool execution, and its signals (truncation scaling, oracle recovery, surface-vs-content metric gap) reproduce across at least two distinct student backbones (§3.6).

Applied to MEM1 (7B RL $\rightarrow$ 3B SFT, with a cross-backbone Llama-1B replication), the protocol yields five findings: (1) multi-objective performance degrades monotonically with no phase transition, setting slope reduction as the distillation target; (2) RL weight changes concentrate $2\text{--}6\times$ supra-random along base principal directions, while SFT recovers only $1.47\times$ —a geometric gap robust across 196 matrices ($p < 1.3 \times 10^{-32}$); (3) DPO fails to close the gap (+1.0pp, power > 0.99 at 5pp), ruling out insufficient training signal; (4) oracle retrieval recovers +47.3pp ($p < 0.0001$), causally confirming that behavioral mismatch accounts for $\sim 86\text{--}88\%$ of the gap, with capacity contributing only $\sim 12\text{--}14\%$ —and even BM25 retrieval recovers +6–12pp, offering an immediately deployable mitigation; (5) the silent-failure signature and behavioral-mismatch diagnosis replicate on Llama-3.2-1B—format-level indicators all at 1.00, action-sequence accuracy 0.606, BLEU 1.16, ROUGE-L 0.21, and oracle retrieval again dropping truncation from 29–86% to 0% while recovering em_partial by ≈ 34 pp at $N = 8$ —evidence that the finding generalises across model families and parameter scales (§3.6).

5.1 Limitations

Scope and capacity isolation. All experiments use one RL teacher (MEM1, 7B) on one domain (multi-objective RAG QA). Student-side we cover two backbones and two scales (Qwen2.5-3B and Llama-3.2-1B, cross-family, $\sim 3\times$ size difference) and observe the same silent-failure signature on both (§3.6); direct replication on other teachers and benchmarks remains future work. The 7B full-rank pooled SFT quantifies the capacity contribution at $\approx 12\text{--}14\%$ of the teacher–student gap (§3.2), but the full-rank vs. LoRA training-method mismatch makes this a best-effort upper bound.

DPO data scarcity and retrieval oracle. DPO uses only 176 matched pairs (0.4% perfect rate at $N = 8$), a structural consequence of multi-objective collapse, not a remediable sampling issue; power exceeds 0.99 for a 5pp effect (floor 2.7pp). Oracle retrieval uses teacher passages and is thus an upper bound, but BM25 retrieval confirms the finding generalizes beyond the oracle setting (+6–12pp, truncation $\rightarrow 0\%$).

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A Training Details

Table 4 summarizes the hyperparameters for all training stages.

	SFT (pooled)	DPO (per-N)
Base model	Qwen2.5-3B	SFT checkpoint
LoRA rank	32	32
LoRA α	64	64
LoRA target	all linear	all linear
Learning rate	2×10^{-4}	2×10^{-5}
Epochs	3	1
Batch size	4	4
Gradient accum.	4	4
Max seq. length	6144	6144
Optimizer	AdamW	AdamW
Weight decay	0.01	0.01
Warmup ratio	0.03	0.1
DPO β	—	0.05
Training samples	1,118 (partial-credit, pooled $N \in \{2, 4, 8\}$)	176 raw pairs (matched to $N = 4$ bottleneck)
Train/test split	hash-based, test_frac = 0.25	

Table 4: Training hyperparameters for SFT and DPO stages. Both stages use LoRA (rank 32) on Qwen2.5-3B. The SFT stage uses partial-credit trajectories (inclusion: total_em ≥ 1); DPO stages use matched-sample preference pairs subsampled to 176 raw pairs.

SFT convergence (3B). The 3B SFT stage converges within 3 epochs with monotonically decreasing training loss (final loss ≈ 0.91). No overfitting is observed on the held-out split.

7B capacity-isolation model. Table 5 summarizes the hyperparameters for the 7B full-rank pooled SFT used in the capacity-isolation experiment (§3.2).

7B SFT (pooled)	
Base model	Qwen2.5-7B
Fine-tuning	Full-rank (no LoRA)
Learning rate	2×10^{-5}
Epochs	2
Batch size	1
Gradient accum.	4 (eff. batch 4)
Max seq. length	6144
Optimizer	AdamW
Precision	bf16
Gradient checkpointing	Yes
Training samples	1,118 (same partial-credit pooled data as 3B SFT)
Training time	19.5 min (560 steps)
Final train loss	0.83

Table 5: Training hyperparameters for the 7B full-rank capacity-isolation SFT. This model uses the same pooled data and inclusion criteria as the 3B LoRA SFT, differing only in model size (7B vs. 3B) and fine-tuning method (full-rank vs. LoRA rank 32). The lower learning rate (2×10^{-5} vs. 2×10^{-4}) and fewer epochs (2 vs. 3) reflect standard practice for full-rank fine-tuning of larger models.

DPO convergence. All three DPO variants ($N \in \{2, 4, 8\}$) train for 1 epoch. Final training losses range from 0.595 (DPO _{$N=2$}) to 0.678 (DPO _{$N=8$}). The rewards/margins are positive but small: 0.207 (DPO _{$N=2$}), 0.100 (DPO _{$N=4$}), 0.056 (DPO _{$N=8$}), consistent with the limited preference signal available from 176 matched pairs. The decreasing margin with N reflects the increasing difficulty of producing contrastive signal at higher objective counts.

B SVD Alignment Per-Layer Distribution

See Figure 3 in the main text for the per-layer SVD alignment boxplot.

C SVD Weight Geometry Details

Definitions. For a base weight matrix $\mathbf{W}_{\text{base}} \in \mathbb{R}^{m \times n}$ with truncated SVD $\mathbf{W}_{\text{base}} = \mathbf{U} \Sigma \mathbf{V}^T$, retain the top- k singular vectors $\mathbf{U}_k, \mathbf{V}_k$. The observed alignment of a fine-tuning delta $\Delta \mathbf{W} = \mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ is

$$\alpha_{\text{obs}}(k) = \frac{\|\mathbf{U}_k^T \Delta \mathbf{W} \mathbf{V}_k\|_F}{\|\Delta \mathbf{W}\|_F},$$

and the alignment ratio is $r(k) = \alpha_{\text{obs}}(k) / \alpha_{\text{rand}}(k)$ where $\alpha_{\text{rand}}(k)$ is the mean projection norm over $n = 100$ Gaussian random matrices matched in shape and entry variance to $\Delta \mathbf{W}$. We decompose into U-side ($\|\mathbf{U}_k^T \Delta \mathbf{W}\|_F / \|\Delta \mathbf{W}\|_F$) and V-side ($\|\Delta \mathbf{W} \mathbf{V}_k\|_F / \|\Delta \mathbf{W}\|_F$) components capturing concentration along base output vs. input directions.

Full alignment ratio table. Across all 196 weight matrices of Qwen2.5-7B (28 layers $\times$ 7 projection types), Table 6 reports $r(k)$ for the three deltas RL-base, SFT-base, and RL-SFT.

	$k=64$	$k=128$	$k=256$	$k=512$	$k=1024$
RL-base	6.48	4.11	2.78	2.01	1.64
SFT-base	2.92	—	1.47	—	—
RL-SFT	3.80	—	1.81	—	—
RL/SFT ratio at matched k :					
$r_{\text{RL}}/r_{\text{SFT}}$	2.22	—	1.89	—	—

Table 6: SVD alignment ratio $r(k)$ for three weight deltas, averaged over 196 matrices. Wilcoxon $p < 1.3 \times 10^{-32}$ at all k . “—” marks k values not computed for SFT/RL-SFT due to compute budget.

Per-matrix dispersion. At $k = 256$ the per-matrix distribution is right-skewed: $r_{\text{RL}} = 2.78 \pm 2.57$, $r_{\text{SFT}} = 1.47 \pm 2.19$ (mean $\pm$ std), with a few outlier layers reaching $r > 10$. Only 2–3 layers show $r_{\text{SFT}} > r_{\text{RL}}$ (most notably layer 1 `mlp.down_proj`); in those cases both ratios remain $r > 2$, so the exceptions are high-alignment for both methods rather than SFT-dominant.

Layer-type decomposition. The alignment gap is not uniform. At $k = 256$, MLP layers show stronger RL-base alignment (3.49 $\times$) than attention layers (2.25 $\times$), while SFT-base alignment is 2.01 $\times$ for MLP and only 1.07 $\times$ for attention—SFT attention deltas are nearly indistinguishable from random. The multi-objective capability encoded in attention weights is therefore the most resistant component to SFT transfer.

U-side / V-side asymmetry. For attention layers, U-side alignment dominates V-side (0.31 vs. 0.12 at $k = 256$): RL modifies how attention writes its output more than how it reads. For MLP layers the pattern reverses (U-side 0.05, V-side 0.11): RL modifies how MLP layers select input features. A successful distillation method would need to replicate not just the magnitude

but the directional structure of RL changes across these layer types.

Same-architecture comparison. Both the RL and SFT models compared here are full-rank fine-tunes of the same Qwen2.5-7B base, so alignment ratios reflect differences in training objective (PPO vs. behavior cloning) rather than dimensional or rank confounds. The distillation experiments in §3 use a separate Qwen2.5-3B / LoRA rank-32 student, reflecting the deployment cost constraint.

D Activation Probing Details

We extract activation differences $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$ at layers $l \in \{0, 3, 7, 11, 15, 19\}$ for $n = 300$ prompts, projecting onto the top- k principal directions ($k \in \{64, 256\}$) of the base model’s weight matrices. Eight feature variants are tested: raw activations $\mathbf{h}^{\text{RL}}$, $\mathbf{h}^{\text{base}}$, their difference $\Delta \mathbf{h}$, and top- k projections thereof.

Linear probes. Binary classification (per-objective EM above/below median) achieves accuracy 0.72, below the majority-class baseline 0.76. Regression probes yield negative R^2 across all 55 (feature, layer, k) combinations.

MLP probes. Two-layer MLPs (hidden dimension 128, ReLU) trained on the same features predict per-objective EM as a continuous target (Spearman ρ). MLP probes lose to linear probes in 47/55 cases (85.5%; paired $t = -8.19$, $p = 4.83 \times 10^{-11}$, Cohen’s $d = -1.105$). Mean ρ : -0.011 (MLP) vs. 0.073 (linear).

Random-feature baseline. We construct 5 random orthonormal bases $\times$ 6 layers $\times$ 2 k values. The maximum $|\rho|$ under random features is 0.224; 52/55 (94.5%) of real linear probe $|\rho|$ values fall within this noise band. The per-objective signal in $\Delta \mathbf{h}$ is indistinguishable from random projections.

E DPO Ablation Details

We train three per- N DPO variants on Qwen2.5-3B (LoRA rank 32), each initialized from the pooled SFT checkpoint and trained on N -specific preference pairs. The matched-sample constraint (176 raw pairs, subsampled to the $N = 4$ bottleneck) is critical for fair cross- N comparison but severely limits data volume: at $N = 8$ only 0.4% of teacher trajectories achieve perfect

scores. This scarcity is a structural consequence of multi-objective collapse.

Model	$N = 2$	$N = 4$	$N = 8$
SFT shared (3B)	0.112	0.084	0.025
SFT full-rank (7B)	0.171	0.147	0.079
DPO $N = 2$	0.128	0.095	0.031
DPO $N = 4$	0.128	0.099	0.026
DPO $N = 8$	0.130	0.082	0.035
RL teacher (7B)	0.59	0.54	0.455

Table 7: Full `em_partial` evaluation matrix (`max_new_tokens = 4096`). Best $N = 8$ DPO: $+1.0\text{pp}$ ($p = 0.163$).

The best result—DPO $_{N=8}$ at eval $N = 8$ —yields $\Delta = +1.0\text{pp}$ ($p = 0.163$, 95% CI $[-0.9, +2.9]\text{pp}$), far below the pre-registered practical-significance threshold of $+5\text{pp}$. A post-hoc power analysis confirms an informative null: with $n = 125$ paired test prompts and observed $\sigma_{\text{diff}} \approx 10.8\text{pp}$, we achieve > 0.99 power at 5pp and 0.80 power at 2.7pp . No systematic N -matching effect is observed: DPO $_{N=2}$ performs comparably to DPO $_{N=8}$ across conditions. Both students retain $< 8\%$ of teacher performance at $N = 8$ (SFT 0.025, DPO $_{N=8}$ 0.035), and neither achieves `em_full` > 0 at $N \geq 4$.

F Extended Related Work

Multi-objective RL for LLMs. Recent work explores multi-objective optimization in LLM alignment: Rewarded Soups (Rame et al., 2024) interpolates weights fine-tuned on diverse rewards, MODPO (Zhou et al., 2024b) extends DPO to multiple preference dimensions, and fine-grained human feedback (Wu et al., 2024) decomposes rewards into per-aspect signals. These methods address *training* with multiple objectives; our work addresses the complementary problem of *distilling* multi-objective RL policies.

Preference learning and DPO. DPO (Rafailov et al., 2023) and extensions (IPO (Gheshlaghi Azar et al., 2024), KTO (Ethayarajh et al., 2024), RAFT++ (Dong et al., 2023)) provide offline alternatives to RLHF. We use DPO as a stronger-than-SFT baseline; its failure ($+1.0\text{pp}$, not significant) suggests behavioral mismatch rather than insufficient signal richness. Following Karl et al. (2024), we

frame this as a diagnostic contribution. 819

G Truncation Confound Analysis 820

An important methodological episode shaped our analysis. An earlier evaluation used `max_new_tokens = 2048`, under which the DPO advantage appeared significant (Table 8). Re-evaluating SFT at 4096 tokens revealed the original significance was largely an artifact.

	em_partial	trunc.	Δ vs SFT	p
<i>SFT at max_new_tokens = 2048 (confounded):</i>				
SFT shared	0.011	93.6%	—	—
DPO $N=8$	0.035	— [†]	+2.4pp	< 0.001
<i>SFT at max_new_tokens = 4096 (corrected):</i>				
SFT shared	0.025	83.2%	—	—
DPO $N=8$	0.035	~84%	+1.0pp	0.163

Table 8: Truncation confound at $N = 8$. Extending SFT’s budget to 4096 tokens lifted `em_partial` from 0.011 to 0.025 and revealed the DPO advantage is not significant. [†]DPO was always evaluated at 4096.

At 2048 tokens, 93.6% of SFT outputs were truncated before the `<answer>` tag; extending to 4096 allowed ~10% more to complete. DPO’s score was unchanged—its apparent advantage came from SFT being disproportionately penalized. A pilot attempt to suppress search tokens via bigram blocking was inconclusive: the model generated tag variants (malformed closing tags, inserted whitespace) that bypassed token-level suppression, suggesting the behavioral pattern is encoded above individual token sequences. This confound is specific to RL-distilled models whose procedural strategies produce verbose outputs; we report it as a methodological caution for future work.

H Full Silent Failure Metric Breakdown

	Format-level	Content-level
Qwen-2.5-3B	≈ 1.00 (all checks)	em_p=0.025
Llama-3.2-1B	≈ 1.00 (all checks)	em_p=0.014

Table 9: Silent failure signature replicates across backbones. Format-level indicators (tool-call syntax, balanced XML tags, non-empty queries) ≈100% for both; content-level `em_partial` near-floor at $N = 8$ (Qwen query Jaccard = 0.95, also near-ceiling). Dispersion analysis in Appendix I.

I Surface Metric Dispersion Analysis

Near-identical cross-backbone point estimates on action-sequence accuracy, BLEU, and ROUGE-L (Table 9) raise a concern that the agreement is coincidental. Per-sample dispersion analysis is deferred to camera-ready pending full access to per-sample surface metrics for both backbones; the point estimates are consistent with floor-level bunching in both backbones given the 125-sample $N = 8$ test split and BLEU values below 2 on the 0–100 scale. Specifically: (i) action-sequence accuracy is dominated by a shared initial “search” action prefix that both students emit deterministically, producing discrete mass at identical values; (ii) BLEU on `sacrebleu`’s 0–100 scale sits near the floor for both students (1.14 Qwen, 1.16 Llama), where noise is small relative to the teacher’s 0.455 `em_partial` signal; and (iii) ROUGE-L at 0.21 reflects shared tool-call markup tokens rather than substantive content overlap. The consistency is therefore a mechanistic consequence of the silent-failure regime, not a point-estimate coincidence, and does not affect any inference in the main text (all causal claims rest on the oracle-recovery contrast, not on surface-metric equality).